Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior software engineer with 6+ years building **scalable, reliable backend systems** and owning them **end to end, from architecture through deployment**, for fully remote teams in the US, Singapore and Uganda. Led a **zero-downtime migration of 20+ production applications** from Parse/MongoDB to Firebase/GCP, reporting to the CTO, with live two-way data sync during cutover and $5,000/month in cloud savings. Built **field tools for NGOs in Uganda**, including a USSD legal-aid service for people without smartphones and a **microservices** case-management backend on Kafka and Kubernetes. Based in Kampala, working in East Africa Time with daily US overlap.

CORE COMPETENCIES

Backend Architecture & APIs | Databases & Cloud Services (GCP, AWS) | Zero-Downtime Migrations | Microservices & Serverless | Deployment Infrastructure (Docker, Kubernetes) | Event-Driven Systems (Pub/Sub, Kafka) | Field Tools for Underserved Users (USSD) | Cross-Functional Ownership

WORK EXPERIENCE

MTailor
Sep 2022 – Present

Software Engineer · US Remote · e-commerce

- Owned the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, from architecture through cutover, reporting directly to the CTO.
- Implemented **real-time two-way sync** between MongoDB and Firestore in Node.js on Google **Pub/Sub**, keeping two live production databases consistent through the migration.
- Moved all services off AWS (S3, EC2, EBS) to GCP, saving **$5,000/month**; migrated file storage from Amazon S3 to GCS with a Python script.
- Shipped product features end to end: Express Shipping (**+$40 revenue per order**) and a 3D visualisation feature (video overlay + ffmpeg) that increased buyer conversion; migrated a Webflow site from Parse to Firebase.

MTailor
Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Built the migration script that read from MongoDB, ran complex processing logic and wrote to Firestore.
- Wrote documentation for fellow engineers on the new Firebase SDKs (Node.js and Python) and trained the Ops team on the new dashboard and backup procedures.

Dr Wealth
Aug 2021 – Jan 2022

Software Engineer · Singapore Remote · fintech

- Extended backend APIs in **TypeScript** on **serverless** Firebase Cloud Functions and Express.js (Heroku) for a real-time stock-price PWA.
- Built ad-hoc jobs over **2M+ Firestore records** to keep customer prices current from Morningstar APIs; added responsive React/Tailwind pages.

Mind2matter
Jan 2022 – Jul 2022

Software Engineer · US Remote · agency

- Built backends for DeFi applications in Node.js with Web3 and delivered React UIs from Figma designs under tight timelines, reporting to the CTO.

CodeBits
Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote · legal tech for NGOs

- Led a team of 4 developers building legal-tech systems for NGOs in Uganda: a case management system for **FIDA Uganda** and a paralegal database for **LASPNET Uganda**.
- Built a **USSD service** so legal-aid providers could reach users on basic phones, communicating with a Node.js backend via **gRPC**.
- Architected a **microservices** backend for FIDA Uganda's case management app using **Apache Kafka, Docker and Kubernetes**.
- Developed a cross-platform mobile app for LASPNET with React Native and Expo (iOS and Android) and Firebase push notifications.

PROJECTS

Real-time Pub/Sub Sync Service

data consistency

Two-way MongoDB↔Firestore sync running under parallel production traffic across 20+ applications during a zero-downtime migration.

Node.js · Google Pub/Sub · MongoDB · Firestore

Healthcare WhatsApp Assistant

Conversational assistant that reads prescription photos and replies to patients in WhatsApp: OpenAI for dialogue, Google Vision for OCR, behind the WhatsApp Business API.

Node.js · OpenAI API · Google Cloud Vision · WhatsApp Business API

EDUCATION

B.Sc. Computer Engineering — Makerere University 2018 – 2022

Kampala, Uganda

SKILLS

Languages: JavaScript, TypeScript, Python, SQL, Go (intermediate), Rust (intermediate)

Backend: Node.js, Express.js, REST APIs, gRPC, Microservices, Serverless (Firebase Cloud Functions)

Data & Messaging: MongoDB, Firestore, Google Pub/Sub, Apache Kafka, NATS Streaming

Cloud / Infra: GCP, Firebase, AWS (S3, EC2, EBS — intermediate), Docker, Kubernetes, Git, Linux

Frontend / Mobile: React, React Native, Tailwind, HTML, CSS, USSD **Spoken:** English, Kiswahili